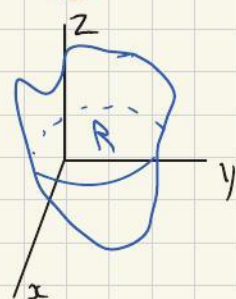


Lec 25: Triple integrals in rectangular and cylindrical coordinates

Triple integrals

$$\iiint_R F dV$$

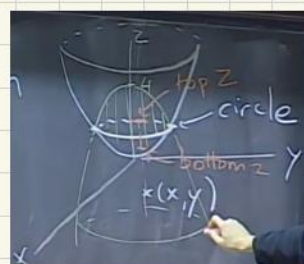
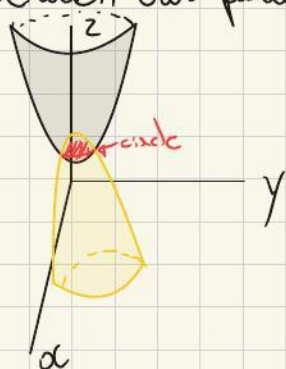
volume element = $dx dy dz$



Example 1: region between two paraboloids: $z = x^2 + y^2$

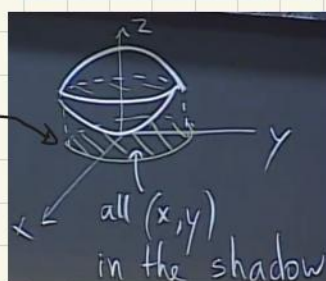
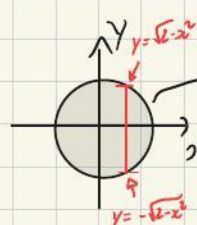
$$z = 4 - x^2 - y^2$$

volume $\iiint dV$



$$\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} dz dy dx$$

top z
bottom z



How to find the shadow in xy-plane?

Wherever $z_{\text{bottom}} < z_{\text{top}}$

$$x^2 + y^2 < 4 - x^2 - y^2$$

$$x^2 + y^2 < 2 = \text{disk of radius } \sqrt{2}$$

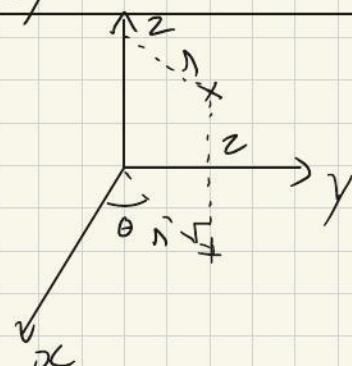
$$\text{Inner: } \int_{x^2+y^2}^{4-x^2-y^2} dz = \left[z \right]_{x^2+y^2}^{4-x^2-y^2} = 4 - 2x^2 - 2y^2$$

$$\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} (4 - 2x^2 - 2y^2) dy dx = \text{switch to polar coordinates}$$

Better: use polar coords instead of x,y

$$\int_0^{2\pi} \int_0^{\sqrt{2}} \int_{x^2+y^2}^{4-x^2-y^2} dz \cdot r dr d\theta \quad (\text{easier to evaluate})$$

cylindrical coordinates (r, θ, z)



θ = angle from the x-axis counterclockwise

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\text{Volume element: } dx dy dz = r dr d\theta dz$$

Applications: * Mass: density $\delta = \frac{\Delta m}{\Delta V}$ $dm = \delta \cdot dV$

$$\text{Mass} = \iiint_R \delta \, dV \quad : \text{if } \delta=1 \text{ then becomes volume}$$

* Average value of $f(x,y,z)$ in R :

$$\bar{f} = \frac{1}{\text{Vol}(R)} \iiint_R f \, dV$$

or with density (weighted average) $\frac{1}{\text{Mass}(R)} \iiint_R f \delta \, dV$

* Center of mass: $(\bar{x}, \bar{y}, \bar{z})$

$$\bar{x} = \frac{1}{\text{Mass}} \iiint_R x \delta \, dV \quad (\text{same for } \bar{y} \text{ and } \bar{z})$$

(In previous example $\bar{x}$ and $\bar{y} = 0$ by symmetry)

* Moment of Inertia: over an axis

$$\iiint_R (\text{distance to axis})^2 \delta \, dV$$

$$I_z = \iiint_R x^2 \delta \, dV = \iiint_R (x^2 + y^2) \delta \, dV$$

moment of inertia over z axis

$$I_x = \iiint_R y^2 + z^2 \delta \, dV$$

$$I_y = \iiint_R x^2 + z^2 \delta \, dV$$

Example 2: I_z of a solid cone between $z=az$ and $z=b$

$\delta=1$

$$I_z = \int_0^b \int_0^{2\pi} \int_0^{z/a} r^2 \cdot r \, dr \, d\theta \, dz = \frac{\pi b^5}{10a^4}$$

Equation of the cone: $r = \frac{z}{a}$



Example 3: setup $\iiint$ for region $z > 1-y$ inside unit ball centered at the origin $x^2 + y^2 + z^2 \leq 1$



$$\int_0^1 \int_{-\sqrt{2y-y^2}}^{\sqrt{2y-y^2}} \int_{1-y}^{\sqrt{1-x^2-y^2}} dz \, dx \, dy$$

When is the plane below sphere?

$$1-y < \sqrt{1-x^2-y^2}$$

$$(1-y)^2 < 1-x^2-y^2$$